

CSCD - GROUP 2 - CAPSTONE PROJECT · FINAL PRESENTATION

Ghost Identity Hunter

Automated OSINT Attribution Platform

Version 2.0 · August 2026

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Multi-Vector OSINT Collection

>>

Graph-Based Identity Correlation

>>

Chain-of-Custody Reporting

Team Members

- 01** C Gunasekaran - Group Lead (Guna)
- 02** Dr. Kanti Singh Sangher
- 03** Rekha Saraswat
- 04** Vindya Narsian
- 05** Soumo Dutta
- 06** Dhayanidhi Rajasekaran (Dhaya) - 2512154

Agenda

- 01** Intro, Team & Scope of Work — Guna (2 min)
- 02** Problem Statement & OSINT Landscape — Rekha (3 min)
- 03** Architecture & Flow Walkthrough — Dhaya (5 min)
- 04** Live Tool Demos — Phone, Maigret, Rest of Tools – Vindya, Dr. Kanti, Soumo (15 min)
- 05** Next Steps & Thanks — Soumo, Guna (2 min)

Case Study Research

(Problem Statement, Analysis & Learnings)

Rekha Saraswat

Business Impact & Objectives

THE PROBLEM

Threat actors construct synthetic identities by compartmentalizing their footprint:

- Burner / VoIP phone numbers (no personal ID required)
- Disposable or forwarding email addresses
- Usernames registered across dozens of platforms
- Stolen or stock profile photographs

Manual OSINT correlation across multiple tools is:

- Time-consuming — hours per investigation
- Error-prone — no quantified confidence
- Inconsistent across analysts
- Missing chain-of-custody audit trail

THE SOLUTION

Ghost Identity Hunter automates the full attribution pipeline:

- ✓ Multi-vector OSINT collection (phone, email, username, image) via BFS orchestration
- ✓ Graph-theoretic correlation links fragmented artifacts into unified persona clusters
- ✓ Quantified confidence scoring per relationship with staleness decay model
- ✓ Generated HTML Report
- ✓ Zero infrastructure cost — local CLI, SQLite, no cloud dependency

Key Components & Technologies

BFS Orchestrator

Python deque · MAX_DEPTH=2 · per-investigation SQLite isolation

Email Intelligence

Holehe (100+ services) · h8mail · HIBP API

Image Intelligence

Pillow (EXIF) · face_recognition (dlib)

Confidence Scorer

Typed link weights · staleness decay · cross-type density metric

Phone Intelligence

phoneinfoga (subprocess) · phonenumbers · E.164 · VoIP detection

Username Enumeration

Sherlock / Maigret · httpx · 400+ platforms · false-positive suppression

Identity Graph

NetworkX DiGraph · Connected components · pyvis interactive visualization

Report & Export

Jinja2 HTML · WeasyPrint PDF · python-stix2 STIX 2.1 bundle

OSINT Tools

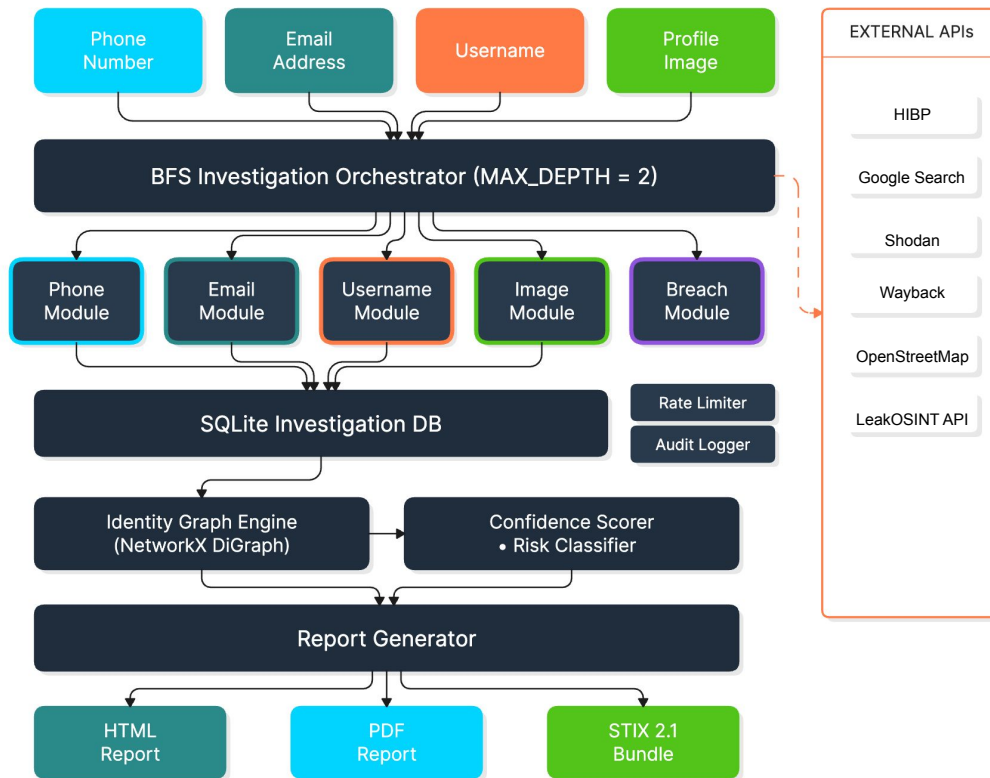
```
$ sherlock elonmusk --print-found --timeout 5 --no-color --no-txt
$ maigret elonmusk --top-sites 150 --timeout 5 --no-progressbar --no-color -J ndjson
$ usufy -n elonmusk -t social -T 32 -e json -o /tmp/osr --avoid_download
$ holehe example@gmail.com --only-used --no-color --no-clear -C
$ theHarvester -d tesla.com -b duckduckgo -f /tmp/report
$ subfinder -d tesla.com -silent -json -timeout 10
$ sublist3r -d tesla.com -n
$ amass enum -passive -d tesla.com
$ whois tesla.com
$ dig +short tesla.com
$ whatweb --color=never --no-errors -a 1 --log-json=/tmp/whatweb.json https://tesla.com
$ nmap -Pn -F -sV --version-light -oX /tmp/scan.xml scanme.nmap.org
$ shodan host 8.8.8.8
$ exiftool -json ~/Pictures/photo.jpg
$ python3 photon.py -u https://tesla.com -o /tmp/photon
```

```
$ python3 sf.py -l 127.0.0.1:5001 (SpiderFoot)
$ recon-ng
$ ghunt email example@gmail.com
$ social-analyzer --username elonmusk
$ python3 metagoofil.py -d tesla.com -t pdf,docx,pptx,xlsx -o /tmp/metagoofil
$ nikto -h https://tesla.com
$ sqlmap -u "http://testphp.vulnweb.com/listproducts.php?cat=1" --batch
$ masscan scanme.nmap.org -p1-1000 --rate 1000
$ curl -L https://tesla.com
$ wget https://tesla.com
$ nslookup tesla.com
$ torbrowser-launcher
$ firefox https://tesla.com (Wappalyzer, Flagfox, UA Switcher)
$ Wayback: curl "https://web.archive.org/cdx/search/cdx?url=tesla.com/*&output=json&limit=15"
```

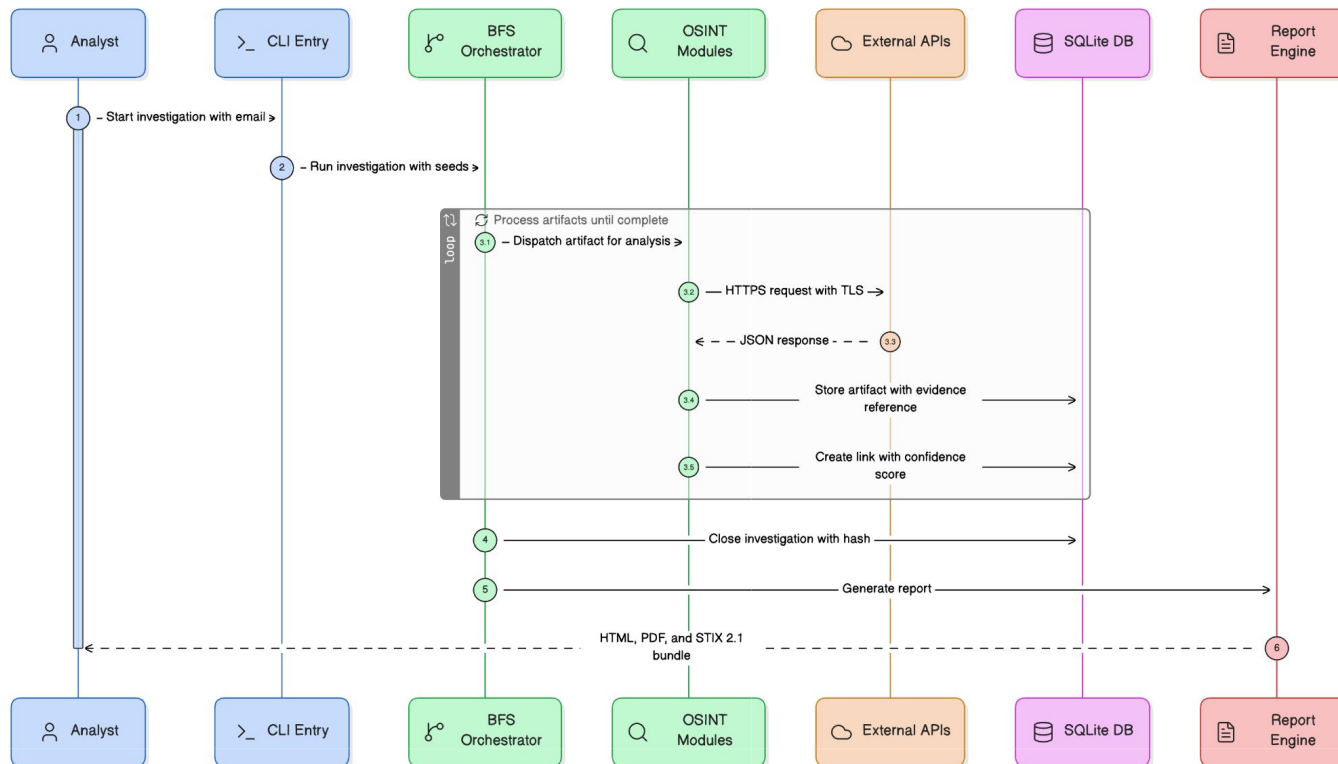
Architectural Deep Dive

Dhaya

Solution Architecture



Sequence Diagram



Sequence Walkthrough — End-to-End Investigation

1

**CLI
Seed & Dispatch**`cli.py investigate` calls `orchestrator.run_investigation()`; seeds load into the BFS queue

2

**BFS
Per-Artifact Processing**

Each artifact hits its OSINT module, external-tool dispatch, and the plugin system in parallel

3

**Discover
Expand the Frontier**

New discovered artifacts are deduped, persisted, and queued for the next BFS depth level

4

**Score
Correlate & Finalize**`analyze_correlation()` builds the identity graph; results are stored in `investigation_metadata`

5

**Report
Auto-Generate Report**`generate_html_report()` / `generate_json_report()` render the finished investigation to reports/

Sequence Walkthrough — How the Report Gets Produced

1

**Read
Load Investigation**

cli.py report reads artifacts, links, platform presences and the audit trail from SQLite

2

**Link
Correlate Identities**

linker.correlate_identities() rebuilds the identity graph and groups connected components

3

**Score
Score & Classify**

compute_identity_risk_score() and classify_risk_level() attach risk to each identity profile

4

**Render
Build Sections & Graph**

Timeline, key findings, risk matrix, heatmap, and an embedded pyvis relationship graph are built

5

**Write
Render & Save**

Jinja2 renders the selected template; HTML is written to reports/{investigation_id}_report.html

Identity Graph & Confidence Scoring

IDENTITY GRAPH (NetworkX DiGraph)

Nodes: Artifact records (phone, email, username, image)

Edges: Directed weighted relationships (link_type + confidence score)

Algorithm:

1. Build DiGraph from DB artifacts + links
2. Find weakly connected components
3. Each component = one persona cluster
4. Score = cross-type link density / 5.0 (capped at 1.0)

Risk classification per cluster:

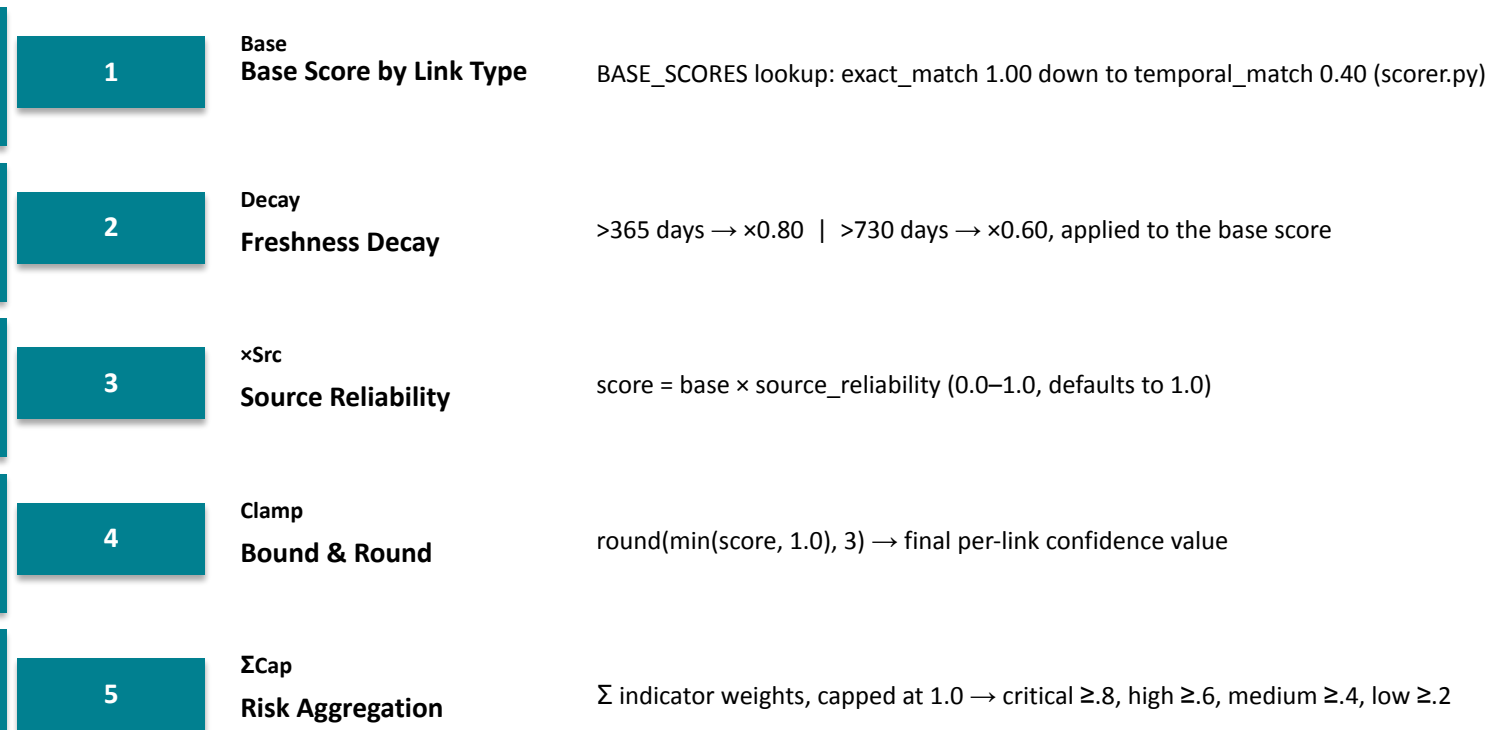
burner_or_voip_phone
disposable_email_address
credential_breach_exposure
stolen_profile_photograph

CONFIDENCE SCORING MODEL

Link Type	Score
exact_match	1.00
registered_with	0.90
breach_linked	0.80
image_match	0.70
username_pattern	0.60
temporal_match	0.40

Staleness Decay: >365 days → ×0.80 | >730 days → ×0.60

Confidence Scoring Algorithm — How It Works



Demo

Vindya, Dr. Kanti & Soumo

Phone Number Lookup — Real Investigation Report

- **PhoneNumber** - country, region, carrier, and number format
- **PhoneValidation** - number is valid and properly formatted and reachable or associated with a particular carrier.
- **LeakOSINT** - searches for publicly exposed information associated with a phone number, such as data appearing in known leaks or databases.

Ghost Identity Hunter - OSINT Investigation Report

Investigation ID: INV-e250ede3 | Title: Investigation INV-e250ede3 | Status: Completed | Generated: 2026-08-08 12:34 UTC

CONFIDENTIAL — INVESTIGATIVE USE ONLY

Case reference: INV-e250ede3 Opened: 2026-08-08T12:34:28 Report generated: 2026-08-08 12:34 UTC Artifacts / links: 11 / 10

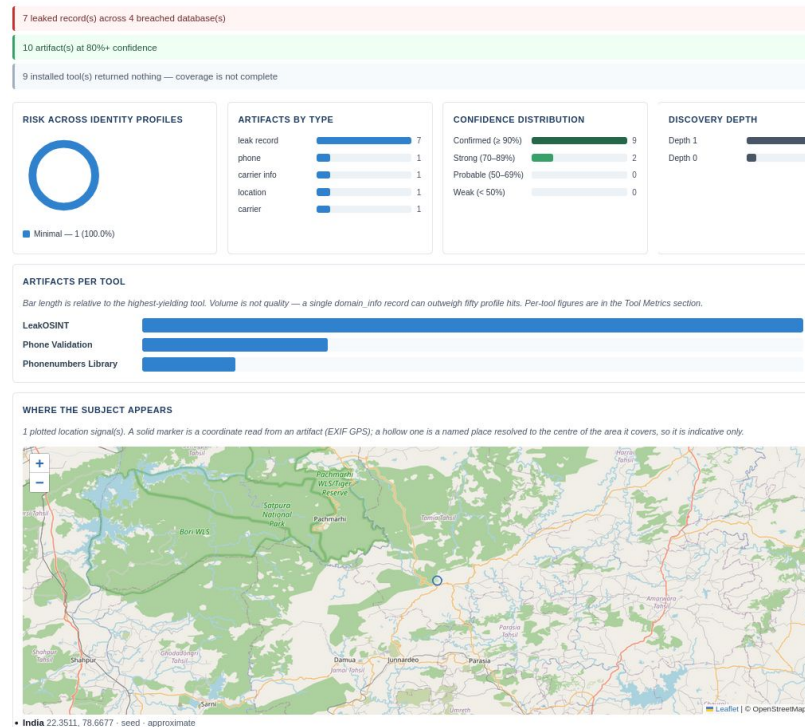
Identity profiles: 1

Handling: Automated analysis — requires analyst review

OSINT Overview

ARTIFACTS	IDENTITY PROFILES	PLATFORM ACCOUNTS	BREACH RECORDS	HIGHEST RISK	MEAN CONFIDENCE
11 10 discovered, 1 seeded	1 10 links between artifacts	0 none found	7 1 leaked rows matched	Minimal 1 profile(s) at this level	89% 3 tool(s) produced output

Investigation overview & dashboard



Evidence basis + PhoneNumber/PhoneValidation/LeakOsint Plugin findings

Maigret Flow — Real Investigation Report

Ghost Identity Hunter - OSINT Investigation Report

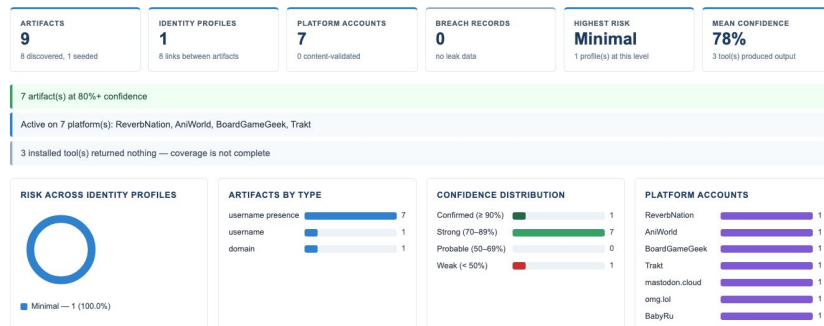
Investigation ID: INV-4579f33e | Title: Demo - Maigret Flow | Status: Completed | Generated: 2026-08-08 11:04 UTC

CONFIDENTIAL — INVESTIGATIVE USE ONLY

Case reference: INV-457933e Opened: 2026-08-08T11:04:12 Report generated: 2026-08-08 11:04 UTC Artifacts / links: 9 / 8 Identity profiles: 0

Handling: Automated analysis — requires analyst review

OSINT Overview



- 3. Tool Metrics

What each OSINT tool actually contributed to this investigation: how many artifacts it produced, of which types, at what average confidence, and how many identity profiles its output reached. Seeded artifacts are excluded, so the totals here count only discovered evidence. Rows marked *derived* are pipeline steps rather than tools — the orchestrator pivoting one artifact into another — and are excluded from the tool count.

3 tool(s) produced 8 artifact(s) across 2 type(s); highest yield was Sherlock with 6. The chart of this distribution is in the OSINT Overview

Per-Tool Breakdown

TOOL	ARTIFACTS	SHARE	AVG. CONFIDENCE	IDENTITIES REACHED	ARTIFACT TYPES
Sherlock	6	75.0%	0.80	1	USERNAME_PRESENCE 6
Google dorks	1	12.5%	0.49	1	DOMAIN 1
Maigret	1	12.5%	0.75	1	USERNAME_PRESENCE 1

Integrated but silent in this run (13 of 15): amass, exiftool, holehe, leakosint, nmap, osrframework, shodan, subfinder, sublist3r, theharvester, wayback-machine, whatweb, whois. A tool is silent when it is n...

Investigation overview — 9 artifacts, 7 platforms

Per-tool breakdown — Sherlock, Google Dorks, Maigret

Next Steps

Soumo

Known Limitations

1	Provider Coverage Gap 4 keyed API providers + 22 local CLI plugins vs. Maltego's 99+ transforms	High
2	No CTI/STIX Export Threat-actor / indicator export (STIX 2.1) is designed but not yet implemented	Medium
3	Concurrency Risks Nested thread pools can oversubscribe; rate-limiter holds its lock during sleep	High
4	No Type-Check Gate mypy is not wired into CI yet, despite type hints being present throughout	Medium
5	Under-tested Modules exports.py (32%) and http_client.py (61%) fall well below the 71% average	Medium
6	Face-Match Perf CPU-bound face_recognition runs on threads, not a ProcessPoolExecutor	Low

Future Roadmap — Next Phases

Next**P1
Concurrency
Hardening**

Fix rate-limiter lock-during-sleep bug; move Google Dorks onto the pooled HTTP session

Next**P2
Performance**

ProcessPoolExecutor for CPU-bound face matching; async I/O for platform fan-out

Next**P3
CTI Export**

Implement STIX 2.1 bundle export — ThreatActor / Indicator / Relationship / Note SDOs

Next**P4
Quality Gate**

Add mypy --strict to CI; raise coverage on exports.py and http_client.py toward 85%

Next**P5
Provider Growth**

Expand keyed-API and plugin coverage toward parity with commercial OSINT platforms

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Questions & Discussion

Ghost Identity Hunter — Thank You

Version 0.1.0 · Capstone Program

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Full documentation in /docs

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Live CLI demo available on request

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Open floor for questions & discussion

Thanks!

Appendix

Maigret— An OSINT tool that searches hundreds of websites for a username and extracts publicly available profile information such as names, bios, locations, profile images, follower counts, and account metadata.

```
$ maigret elonmusk --top-sites 150 --timeout 5 --no-progressbar --no-color --no-recursion -J ndjson -fo /tmp/maigret
```

```
[+] MAIGRET - collect a dossier by username from 3000+ sites
[♥] Support Maigret - sites database & development
[♥] Donate on Patreon: https://www.patreon.com/soxoj
[+] Using sites database: /home/cscd/.local/share/pipx/venvs/maigret/lib/python3.13/site-packages/maigret/resources/data.json (3187 sites)
[-] Starting a search on top 160 sites from the Maigret database...
[!] You can run search by full list of sites with flag '-a'
[*] Checking username elonmusk on:
[+] SoundCloud: https://soundcloud.com/elonmusk
  -uid: 231728889
  -name: rapgame Elon Musk
  -username: rapgame Elon Musk
  -following_count: 47
  -follower_count: 9
  -is_verified: False
  -image: https://i1.sndcdn.com/avatars-000229282171-i2n8i0-large.jpg
  -location: Mars
  -bio: beats to teslacoin@yahoo.com
```

```

already platinum on Mars
└─ extractor: SoundCloud
[+] Instagram: https://www.instagram.com/elonmusk/
├─ username: elonmusk
├─ id: 27120772740
├─ image: https://instagram.fmaa2-3.fna.fbcdn.net/v/t51.82787-19/686366405_18098858740908741_2994877721432835691_n.jpg?stp=dst-jpg_s320x320_tt6&efg=eyJ2ZW5jb2RlX3RhZyI6InByb2ZpbGVfG1JmRqYW5nby40MDAuYzI1fQ&_nc_cat=109&_nc_oc=Q6cZ2gE5qYUqkiQw15-mjq9G02yDtiD-Yu0CKpQk3J0aaZg7JewLxcx__Um_mo_448uOwo&_nc_ohc=n7yzPwK9fsMQ7kNvwEL5wyN&_nc_gid=24_Ff0jVXWLpqU81WJ3w&edm=ADQ1c0wBAAAA&ccb=7-5&oh=00_AQGo2Lth8VY-za1-4d6uvWlr-D0kgueGzgvsJGwgn12W_PA&oe=6A7CD85B&_nc_sid=8b3546
├─ facebook_uid: 17841427262200096
├─ is_business: False
├─ is_joined_recently: False
├─ is_private: False
├─ is_verified: True
├─ follower_count: 144326
├─ following_count: 0
└─ extractor: Instagram GraphQL
[+] Spotify: https://open.spotify.com/user/elonmusk
[+] GitHub: https://github.com/elonmusk
├─ uid: 4688214
└─ username: elonmusk

```

Usufy— Part of the OSRFramework suite that enumerates social media account presences for a target username across various platforms.

```
|$ usufy -n elonmusk -t social -T 32 -e json -o /tmp/osr --avoid_download
```

[illegible]

1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34 35 36 37 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66 67 68 69 70 71 72 73 74 75 76 77 78 79 80 81 82 83 84 85 86 87 88 89 90 91 92 93 94 95 96 97 98 99 100 101 102 103 104 105 106 107 108 109 110 111 112 113 114 115 116 117 118 119 120 121 122 123 124 125 126 127 128 129 130 131 132 133 134 135 136 137 138 139 140 141 142 143 144 145 146 147 148 149 150 151 152 153 154 155 156 157 158 159 160 161 162 163 164 165 166 167 168 169 170 171 172 173 174 175 176 177 178 179 180 181 182 183 184 185 186 187 188 189 190 191 192 193 194 195 196 197 198 199 200 201 202 203 204 205 206 207 208 209 210 211 212 213 214 215 216 217 218 219 220 221 222 223 224 225 226 227 228 229 230 231 232 233 234 235 236 237 238 239 240 241 242 243 244 245 246 247 248 249 250 251 252 253 254 255 256 257 258 259 260 261 262 263 264 265 266 267 268 269 270 271 272 273 274 275 276 277 278 279 280 281 282 283 284 285 286 287 288 289 290 291 292 293 294 295 296 297 298 299 300 301 302 303 304 305 306 307 308 309 310 311 312 313 314 315 316 317 318 319 320 321 322 323 324 325 326 327 328 329 330 331 332 333 334 335 336 337 338 339 340 341 342 343 344 345 346 347 348 349 350 351 352 353 354 355 356 357 358 359 360 361 362 363 364 365 366 367 368 369 370 371 372 373 374 375 376 377 378 379 380 381 382 383 384 385 386 387 388 389 390 391 392 393 394 395 396 397 398 399 400 401 402 403 404 405 406 407 408 409 410 411 412 413 414 415 416 417 418 419 420 421 422 423 424 425 426 427 428 429 430 431 432 433 434 435 436 437 438 439 440 441 442 443 444 445 446 447 448 449 450 451 452 453 454 455 456 457 458 459 460 461 462 463 464 465 466 467 468 469 470 471 472 473 474 475 476 477 478 479 480 481 482 483 484 485 486 487 488 489 490 491 492 493 494 495 496 497 498 499 500 501 502 503 504 505 506 507 508 509 510 511 512 513 514 515 516 517 518 519 520 521 522 523 524 525 526 527 528 529 530 531 532 533 534 535 536 537 538 539 540 541 542 543 544 545 546 547 548 549 550 551 552 553 554 555 556 557 558 559 560 561 562 563 564 565 566 567 568 569 570 571 572 573 574 575 576 577 578 579 580 581 582 583 584 585 586 587 588 589 590 591 592 593 594 595 596 597 598 599 600 601 602 603 604 605 606 607 608 609 610 611 612 613 614 615 616 617 618 619 620 621 622 623 624 625 626 627 628 629 630 631 632 633 634 635 636 637 638 639 640 641 642 643 644 645 646 647 648 649 650 651 652 653 654 655 656 657 658 659 660 661 662 663 664 665 666 667 668 669 670 671 672 673 674 675 676 677 678 679 680 681 682 683 684 685 686 687 688 689 690 691 692 693 694 695 696 697 698 699 700 701 702 703 704 705 706 707 708 709 710 711 712 713 714 715 716 717 718 719 720 721 722 723 724 725 726 727 728 729 730 731 732 733 734 735 736 737 738 739 740 741 742 743 744 745 746 747 748 749 750 751 752 753 754 755 756 757 758 759 760 761 762 763 764 765 766 767 768 769 770 771 772 773 774 775 776 777 778 779 780 781 782 783 784 785 786 787 788 789 790 791 792 793 794 795 796 797 798 799 800 801 802 803 804 805 806 807 808 809 810 811 812 813 814 815 816 817 818 819 820 821 822 823 824 825 826 827 828 829 830 831 832 833 834 835 836 837 838 839 840 841 842 843 844 845 846 847 848 849 850 851 852 853 854 855 856 857 858 859 860 861 862 863 864 865 866 867 868 869 870 871 872 873 874 875 876 877 878 879 880 881 882 883 884 885 886 887 888 889 890 891 892 893 894 895 896 897 898 899 900 901 902 903 904 905 906 907 908 909 910 911 912 913 914 915 916 917 918 919 920 921 922 923 924 925 926 927 928 929 930 931 932 933 934 935 936 937 938 939 940 941 942 943 944 945 946 947 948 949 950 951 952 953 954 955 956 957 958 959 960 961 962 963 964 965 966 967 968 969 970 971 972 973 974 975 976 977 978 979 980 981 982 983 984 985 986 987 988 989 990 991 992 993 994 995 996 997 998 999 1000 1001 1002 1003 1004 1005 1006 1007 1008 1009 1010 1011 1012 1013 1014 1015 1016 1017 1018 1019 1020 1021 1022 1023 1024 1025 1026 1027 1028 1029 1030 1031 1032 1033 1034 1035 1036 1037 1038 1039 104

Coded with ❤ by [Yaiza Rubio](#) & [Félix Brezo](#)

Version 2.0 · Capstone Program · June 2026

SubFinder— A fast, passive subdomain discovery tool that extracts valid subdomains using various public resources and APIs.

```
L$ subfinder -d tesla.com -silent -json -timeout 10

{"host":"pop.akamai-apigateway-suppliers.tesla.com","input":"tesla.com","source":"submd"}
{"host":"public.digitalassets-shop.tesla.com","input":"tesla.com","source":"submd"}
{"host":"stream.hermes-api.prd.na.vn.cloud.tesla.com","input":"tesla.com","source":"submd"}
{"host":"media.extgithub.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"mfs-supplier-uat.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"dev.assistant-api.prd.usw2.vn.cloud.tesla.com","input":"tesla.com","source":"submd"}
{"host":"vehicle-files.prd.vn.cloud.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"www-static-dev.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"invest.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"dev.fleet-apf-prd.tesla.com","input":"tesla.com","source":"submd"}
{"host":"lyncdiscover.contractors.contractors.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"alpha.epuca-prd.usw.vn.cloud.tesla.com","input":"tesla.com","source":"submd"}
{"host":"cdn.location-services-prd.tesla.com","input":"tesla.com","source":"submd"}
{"host":"forum.digitalcontent.tesla.com","input":"tesla.com","source":"submd"}
{"host":"usw2.vn.cloud.tesla.com","input":"tesla.com","source":"submd"}
{"host":"signaling-robotics.eng.vn.cloud.tesla.com","input":"tesla.com","source":"shodanct"}
{"host":"staging.assistant-api.prd.usw2.vn.cloud.tesla.com","input":"tesla.com","source":"submd"}
{"host":"v2.digitalassets-accounts.tesla.com","input":"tesla.com","source":"submd"}
{"host":"codeload.github.tesla.com","input":"tesla.com","source":"shodanct"}
{"host":"exp-zcp-cloud-na-itmfg.tesla.com","input":"tesla.com","source":"shodanct"}
{"host":"go.apigateway-logging.tesla.com","input":"tesla.com","source":"submd"}
{"host":"edr-eu.tesla.com","input":"tesla.com","source":"thc"}
{"host":"sin05-gpgw1.tesla.com","input":"tesla.com","source":"shodanct"}
{"host":"sso.tesla.com","input":"tesla.com","source":"shodanct"}
{"host":"autobidder-eng.powerhub.energy.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"kronosdb.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"model3.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"origin-apps.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"origin-bolt.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"dns1.cua-chat-ui.tesla.com","input":"tesla.com","source":"submd"}
{"host":"ny.tesla.com","input":"tesla.com","source":"shodanct"}
{"host":"tvs.tesla.com","input":"tesla.com","source":"shodanct"}
{"host":"cnvpn1.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"o4.ptr1867.tesla.com","input":"tesla.com","source":"anubis"}
{"host":"gcp.euw1.vn.cloud.tesla.com","input":"tesla.com","source":"submd"}
{"host":"my.assistant-api.prd.na.vn.cloud.tesla.com","input":"tesla.com","source":"submd"}
{"host":"npuv-ap.prd.usw2.vn.cloud.tesla.com","input":"tesla.com","source":"submd"}
{"host":"prometheus.cua-help-me-charge-ui.tesla.com","input":"tesla.com","source":"submd"}
```

SubList3r— Enumerates subdomains of a target domain using OSINT sources like Google, Yahoo, Bing, and SSL certificates.

```
(.venv)-(cscd@ kali1)-[~/gih]
$ sublist3r -d tesla.com -n
```



Coded By Ahmed Aboul-Ela - @aboul3la

```
[~] Enumerating subdomains now for tesla.com
[~] Searching now in Baidu..
[~] Searching now in Yahoo..
[~] Searching now in Google..
[~] Searching now in Bing..
[~] Searching now in Ask..
[~] Searching now in Netcraft..
[~] Searching now in DNSdumpster..
[~] Searching now in Virustotal..
[~] Searching now in ThreatCrowd..
[~] Searching now in SSL Certificates..
[~] Searching now in PassiveDNS..
[!] Error: Virustotal probably now is blocking our requests
[!] DNSDumpster module failed: Could not find CSRF token on DNSDumpster page
```

Whatweb— Analyzes web applications to identify server technologies, CMS platforms, embedded scripts, and HTTP headers.

```
(cscd@kali1)-[~]
$ whatweb --color=never --no-errors -a 1 --open-timeout 10 --read-timeout 20 --max-threads 5 --log-json=/tmp/whatweb.json https://tesla.com

https://tesla.com [403 Forbidden] Akamai-Global-Host, Country[EUROPEAN UNION][EU], HTTPServer[AkamaiGHost], IP[2.18.55.207], Strict-Transport-Security[max-age=15768000], Title[Access Denied], UncommonHeaders[x-reference-error,x-ak-cache,permissions-policy]

(cscd@kali1)-[~]
$ whatweb --color=never --no-errors -a 1 --open-timeout 10 --read-timeout 20 --max-threads 5 --log-json=/tmp/whatweb.json https://www.tesla.com

https://www.tesla.com [403 Forbidden] Akamai-Global-Host, Country[INDIA][IN], HTTPServer[AkamaiGHost], IP[49.44.104.70], Strict-Transport-Security[max-age=15768000], Title[Access Denied], UncommonHeaders[x-reference-error,alt-svc,x-ak-cache,permissions-policy]

(cscd@kali1)-[~]
$ whatweb --color=never --no-errors -a 1 --open-timeout 10 --read-timeout 20 --max-threads 5 --log-json=/tmp/whatweb.json https://www.google.com

https://www.google.com [200 OK] Cookies[AEC,NID,__Secure-STRP], Country[UNITED STATES][US], HTML5, HTTPServer[gws], HttpOnly[AEC,NID], IP[142.251.151.119], Script, Title[Google], UncommonHeaders[content-security-policy-report-only,accept-ch,alt-svc], X-Frame-Options[SAMEORIGIN], X-XSS-Protection[0]

(cscd@kali1)-[~]
$ whatweb --color=never --no-errors -a 1 --open-timeout 10 --read-timeout 20 --max-threads 5 --log-json=/tmp/whatweb.json https://www.samsung.com

https://www.samsung.com [403 Forbidden] Cookies[country_codes,country_region,device_type,s_fpid], Country[UNITED STATES][US], HttpOnly[s_fpid], IP[23.41.120.67], Title[Access Denied], UncommonHeaders[server-timing,alt-svc]

(cscd@kali1)-[~]
$ whatweb --color=never --no-errors -a 1 --open-timeout 10 --read-timeout 20 --max-threads 5 --log-json=/tmp/whatweb.json https://www.verizon.com

https://www.verizon.com [403 Forbidden] Country[UNITED STATES][US], Email[VZGTSASAPMonitoring@VerizonWireless.com,VZGTSASAPMonitoring@VerizonWireless.com%0A ], IP[23.211.60.210], Strict-Transport-Security[max-age=31536000 ; includeSubDomains ; preload], Title[403 Forbidden], UncommonHeaders[cdn-requestid,akamai-grn]

(cscd@kali1)-[~]
$ whatweb --color=never --no-errors -a 1 --open-timeout 10 --read-timeout 20 --max-threads 5 --log-json=/tmp/whatweb.json https://www.microsoft.com

https://www.microsoft.com [403 Forbidden] Country[UNITED STATES][US], HTML5, HTTPServer[AkamaiNetStorage], IP[184.28.110.44], Strict-Transport-Security[max-age=31536000 ; includeSubDomains], Title[Microsoft]
```

NMap— A network mapper that performs fast port scanning and lightweight service/version detection against a host.

```
nmap -Pn -F -sV --version-light -oX /tmp/scan.xml scanme.nmap.org

Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 07:26 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.37s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 92 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp   filtered  msrpc
139/tcp   filtered  netbios-ssn
179/tcp   filtered  bgp
445/tcp   filtered  microsoft-ds
646/tcp   filtered  ldap
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.05 seconds
Starting Nmap 7.99 ( https://nmap.org ) at 2026-08-08 07:26 -0400
Nmap scan report for scanme.nmap.org (45.33.32.156)
Host is up (0.26s latency).
Other addresses for scanme.nmap.org (not scanned): 2600:3c01::f03c:91ff:fe18:bb2f
Not shown: 92 closed tcp ports (reset)
PORT      STATE      SERVICE      VERSION
22/tcp    open      ssh          OpenSSH 6.6.1p1 Ubuntu 2ubuntu2.13 (Ubuntu Linux; protocol 2.0)
25/tcp    filtered  smtp
80/tcp    open      http         Apache httpd 2.4.7 ((Ubuntu))
135/tcp   filtered  msrpc
139/tcp   filtered  netbios-ssn
179/tcp   filtered  bgp
445/tcp   filtered  microsoft-ds
646/tcp   filtered  ldap
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 10.59 seconds
```

Shodan— An Internet-connected device search engine used via CLI to fetch open port, geolocation, and banner details for an IP.

```
[L$ shodan host 8.8.8.8
/home/cscd/.local/share/pipx/venvs/shodan/lib/python3.13/site-packages/shodan/__main__.py:33: UserWarning: pkg_resources is deprecated as an API. See https://setuptools.pypa.io/en/latest/pkg_resources.html. The pkg_resources package is slated for removal as early as 2025-11-30. Refrain from using this package or pin to Setuptools<81.
import pkg_resources
8.8.8.8
Hostnames:      dns.google;search1api-na1.emarketer.com
City:           Mountain View
Country:        United States
Organization:   Google LLC
Updated:        2026-08-08T04:43:44.361908
Number of open ports:  2

Ports:
53/tcp
53/udp
443/tcp
|-- HTTP title: Google Public DNS
|-- Cert Issuer: C=US, CN=WR2, O=Google Trust Services
|-- Cert Subject: CN=dns.google
|-- SSL Versions: -SSLv2, -SSLv3, -TLSv1, -TLSv1.1, TLSv1.2, TLSv1.3
```

Social Analyzer— Analyzes user profiles and searches across 1,000+ social media networks using detection heuristics.

```
L$ social-analyzer --username elonmusk
```

```
[init] Detections are updated very often, make sure to get the most up-to-date ones
[init] languages.json looks good!
[init] sites.json looks good!
[init] languages.json & sites.json loaded successfully
[Init] Selected websites: 999
[Info] username: elonmusk
[Checking] adultism.com
[Checking] akniga.org
[Checking] 7cups.com
[Checking] airbit.com
[Checking] 8tracks.com
[Checking] 24.wikia.com
[Waiting to retry] adultism.com
Some characters could not be decoded, and were replaced with REPLACEMENT CHARACTER.
Some characters could not be decoded, and were replaced with REPLACEMENT CHARACTER.
[Checking] 9gag.com
[Checking] airlinepilot.life
[Checking] akbrny.com
[Checking] 500px.com
[Checking] adore.one
[Checking] about.me
[Checking] aahachat.org
[Checking] admireme.vip
[Checking] aliveshoes.com
[Checking] airliners.net
[Checking] alik.cz
[Checking] alimero.ru
[Checking] allmylinks.com
[Checking] algowiki-project.org
[Checking] alleywatch.com
[Checking] aminoapps.com
[Waiting to retry] aminoapps.com
[Checking] anilist.co
[Checking] anar.biz
[Checking] alltrails.com
[Checking] allthingsworn.com
[Checking] angel.co
[Checking] anobii.com
[Checking] anphabe.com
[Checking] apclips.com
[Checking] apne.co
```

SQLMap – An automated penetration testing tool used to detect and exploit SQL injection vulnerabilities in web applications.

```
$ sqlmap -u "http://testphp.vulnweb.com/listproducts.php?cat=1" --batch
```



[!] legal disclaimer: Usage of sqlmap for attacking targets without prior mutual consent is illegal. It is the end user's responsibility to obey all applicable local, state and federal laws. Developers assume no liability and are not responsible for any misuse or damage caused by this program

```
[*] starting @ 07:38:58 /2026-08-08/
```

```
[07:38:58] [INFO] testing connection to the target URL
```

```
^C
```

```
[*] ending @ 07:39:10 /2026-08-08/
```

SQLite Investigation Schema

investigations

investigation_id TEXT PK
title TEXT
status TEXT CHECK(...)
created_at DATETIME
completed_at DATETIME
db_sha256 TEXT

artifact_links

link_id TEXT PK
source_artifact TEXT FK
target_artifact TEXT FK
link_type TEXT
confidence REAL CHECK(0-1)
evidence TEXT (JSON)

artifacts

artifact_id TEXT PK
investigation_id TEXT FK
artifact_type TEXT CHECK
value TEXT
source_module TEXT
confidence REAL CHECK(0-1)
raw_evidence_ref TEXT

platform_presence

presence_id TEXT PK
platform_name TEXT
profile_url TEXT
username TEXT
is_verified BOOLEAN
raw_evidence_ref TEXT